L3

APPLY GENERAL PHYSICS

MODULE CODE: GENPHY301

LEARNING UNITS

Unit1. Describe basic measurements in physics

Unit2. Describe motivation in one dimension

Unit3. Analyze motivation in two dimensions

Unit4. Demonstrate electrostatic phenomena

Unit5. Apply geometric optics

Unit6. Characterize sources of energy in the world

Physics is the natural science that studies matter, its fundamental constituents, its motion and behavior through space and time, and the related entities of energy and force**.**

Physics is one of the most fundamental scientific disciplines, and its main goal is to understand how the universe behaves.

Advances in physics often enable advances in new technologies. For example, advances in the understanding of electromagnetism, solid-state physics, and nuclear physics led directly to the development of new products that have dramatically transformed modern-day society, such as television, computers, domestic appliances, and nuclear weapons; advances in thermodynamics led to the development of industrialization; and advances in mechanics inspired the development of calculus.

UNIT1. DESCRIBE BASIC MEASUREMENTS IN PHYSICS

1.1 Introduction to basic measurement in physics

A quantity may be defined as any observable property or process in nature with which a number may be associated. This number is obtained by the operation of measurements. The number may be obtained directly by a single measurement or indirectly, say for example, by multiplying together two numbers obtained in separate operations of measurement.

Fundamental quantities are those quantities that are not defined in terms of other quantities. In physics there are 7 fundamental quantities of measurements namely **length, mass, time, temperature, electric current**, amount of substance and luminous intensity.

SI units and symbols

In order to measure any quantity, a standard unit (base unit) of reference is chosen. This system is called the International System of Units (**SI).**

Fundamental quantity	SI Unit	Symbol
1. Length	Metre	m
2. Mass	Kilogram	kg
3. Time	Second	s
4. Temperature	Kelvin	K
5. Electric current	Ampere	A
6. Amount of substance	Moles	mol
7. Luminous Intensity	Candela	cd

1.2 Meaning of fundamental physical quantity

Fundamental physical quantities are the basic, independent quantities that form the foundation for other derived quantities in physics.

● **Mass**: The amount of matter in an object, typically measured in kilograms (kg). It is a measure of an object's resistance to acceleration.

● **Length**: The measurement of distance between two points, typically measured in meters (m).

● **Time**: The duration or interval over which events occur, measured in seconds (s).

1.3 Meaning of delivered physical quantity

Derived physical quantities are quantities that are calculated from two or more fundamental physical quantities, often through mathematical relationships.

● **Volume:** The amount of space occupied by an object, usually measured in cubic units (e.g., cubic meters).

● **Weight:** The force exerted on an object due to gravity, calculated as mass times gravitational acceleration (measured in newton's, N).

● **Density:** The mass per unit volume of a substance, expressed as mass divided by volume (kg/m^3).

● **Area:** The amount of surface an object covers, measured in square units (e.g., square meters, m^2).

● **Force:** A push or pull acting on an object, defined as mass times acceleration (measured in newton, N).

1.4 International system of units (SI) and metric prefixes in everyday use

The International System of Units (SI) is the modern form of the metric system used globally for consistency in measurement. It includes base units for fundamental physical quantities and prefixes that denote powers of ten, making large or small numbers easier to express.

SI Base Units

Quantity	Unit Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Amount of Substance	mole	mol
Luminous Intensity	candela	cd

Metric prefixes are symbols added to the beginning of unit names to represent powers of ten, making it easier to express very large or very small quantities. They help simplify measurements by scaling the base unit up or down by factors of ten.

For example:

The prefix **"kilo-"** (symbol: k) means 1,000 times the base unit (e.g., 1 kilometer = 1,000 meters).

The prefix **"milli-"** (symbol: m) means 1/1,000 (or 0.001) of the base unit (e.g., 1 milligram = 0.001 grams).

Metric Prefixes

Prefix Name	Symbol	Factor	Example
kilo	k	10^3 (1,000)	1 kilometer (1,000 m)
hecto	h	10^2 (100)	1 hectogram (100 g)
deca	da	10^1 (10)	1 decameter (10 m)
deci	d	10^{-1} (0.1)	1 decimeter (0.1 m)
centi	c	10^{-2} (0.01)	1 centimeter (0.01 m)
milli	m	10^{-3} (0.001)	1 milligram (0.001 g)
micro	μ	10^{-6}	1 micrometer (0.000001 m)
nano	n	10^{-9}	1 nanometer (0.000000001 m)

1.5 Dimension analysis

Dimensional analysis is a method in physics used to understand the relationships between different physical quantities by expressing them in terms of their basic dimensions, such as mass, length, and time.

1.5.1 Definition of Dimensions of Physical Quantities

Dimensions refer to the powers to which the fundamental units of mass (M), length (L), and time (T) must be raised to represent a physical quantity. Physical quantities can be expressed in terms of basic dimensions, which help in verifying equations, conversions, and understanding units.

Assigning Dimensions to Physical Quantities

Each physical quantity is assigned dimensions based on how it relates to the fundamental physical quantities of mass (M), length (L), time (T), etc. For example:

The seven fundamental quantities their dimensions.

Fundamental Quantity	Dimension
Length	L
Mass	M
Time	T
Temperature	K

Electric Current	A
Luminous Intensity	Cd
Amount of substance	mol

Let us consider a physical quantity **Q** which depends on base quantities like length, mass, time, electric current, the amount of substance and temperature, when they are raised to powers a, b, c, d, e, and f. Then dimensions of physical quantity Q can be given as:

$[Q] = [L^a M^b T^c A^d mol^e K^f]$

It is mandatory for us to use [ ] in order to write dimension of a physical quantity. In real life, everything is written in terms of dimensions of mass, length and time. Look out few examples given below:

1. The volume of a solid is given is the product of length, breadth and its height. Its dimension is given as:

Volume = Length × Breadth × Height

Volume = [L] × [L] × [L] (as length, breadth and height are lengths)

Volume = $[L]^3$

As volume is dependent on mass and time, the powers of time and mass will be zero while expressing its dimensions i.e. $[M]^0$ and $[T]^0$

The final dimension of volume will be $[M]^0[L]^3[T]^0 = [M^0L^3T]$

2. In a similar manner, dimensions of area will be $[M]^0[L]^2[T]^0$

3. Speed of an object is distance covered by it in specific time and is given as:

Speed = Distance/Time

Dimension of Distance = [L]

Dimension of Time = [T]

Dimension of Speed = [L]/[T]

$[Speed] = [L][T]^{-1} = [LT^{-1}] = [M^0LT^-1]$

4. Acceleration of a body is defined as rate of change of velocity with respect to time, its dimensions are given as:

Acceleration = Velocity / Time

Dimension of velocity = $[LT^{-1}]$

Dimension of time = [T]

Dimension of acceleration will be = $[LT^{-1}]/[T]$

[Acceleration] = $[LT^{-2}] = [M^0LT^{-2}]$

5. Density of a body is defined as mass per unit volume, and its dimension are given as:

Density = Mass / Volume

Dimension of mass = [M]

Dimension of volume = $[L^3]$

Dimension of density will be = $[M] / [L^3]$

[Density] = $[ML^{-3}]$ or $[ML^{-3}T^0]$

6. Force applied on a body is the product of acceleration and mass of the body

Force = Mass × Acceleration

Dimension of Mass = [M]

Dimension of Acceleration = $[LT^{-2}]$

Dimension of Force will be = $[M] \times [LT^{-2}]$

[Force] = $[MLT^{-2}]$

1.5.2. Rules for writing dimensions of a physical quantity

We follow certain rules while expression a physical quantity in terms of dimensions, they are as follows:

- Dimensions are always enclosed in [ ] brackets
- If the body is independent of any fundamental quantity, we take its power to be 0
- When the dimensions are simplified we put all the fundamental quantities with their respective power in single [] brackets, for example as in velocity we write $[L][T]^{-1}$ as $[LT^{-1}]$
- We always try to get derived quantities in terms of fundamental quantities while writing a dimension.
- Laws of exponents are used while writing dimension of physical quantity so basic requirement is a must thing

- If the dimension is written as it is we take its power to be 1, which is an understood thing
- Plane angle and Solid angle are dimensionless quantity, that is they are independent of fundamental quantities
  - Force, [F] = $[MLT^{-2}]$
  - Velocity. [v] = $[LT^{-1}]$
  - Charge, (q) = [AT]
  - Specific heat, (s) = $[L^2T^2K^{-1}]$
  - Gas constant, [R] = $[ML^2T^{-2}K^{-1}\ mol^{-1}]$

1.5.3. Benefits of dimensions

Before writing dimensions of a physical quantity, it is must know a thing to understand why do we need dimensions and what are benefits of writing a physical quantity. Benefits of describing a physical quantity are as follows:

- Describing dimensions help in understanding the relation between physical quantities and its dependence on base or fundamental quantities, that is, how dimensions of a body rely on mass, time, length, temperature etc.
- Dimensions are used in dimension analysis, where we use them to convert and interchange units
- Dimensions are used in predicting unknown formulae by just studying how a certain body depends on base quantities and up to which extent
- It makes measurement and study of physical quantities easier
- We are able to identify or observe a quantity just because of its dimensions

Dimensions define objects and their existence

1.5.4. Limitation of dimensions.

Besides being a useful quantity, there are many limitations of dimensions, which are as follows:

- ➢ Dimensions can't be used for trigonometric, logarithmic and exponential functions which have no dimension.
- ➢ Dimensions never define exact form of a relation

- We can't find values of certain constants in physical relations with the help of dimensions
- It does not give us any information about the dimensional constants in the formula.
- We cannot find the exact sign of plus or minus connecting two or more terms in relation.
- A dimensionally correct equation may not be the correct equation always

1.6 Sources of errors in measurement of physical quantities

A measurement is an observation that has a numerical value and unit. When you measure an object, you compare it with a standard unit. measurement must be expressed by a number and a unit.

Standard measurement is an exact quantity that people agree on to be used for comparison or as a reference to measure other quantities. We have three kinds of standards: International standard, Regional standard and National standard.

1.6.1. Experimental Errors

Experimental error is the difference between a measured value and the true or accepted value of the quantity being measured. It quantifies how much a measurement deviates from the actual result due to limitations in the experiment

Types of Experimental Error.

1. **Systematic Error**: These are consistent errors in measurements. They happen due to flaws in equipment or methods. These errors can be corrected once identified.
2. **Random Error**: These errors vary unpredictably. They occur due to unknown factors or changes in the environment. These can't be fully eliminated but can be reduced with more measurements.
3. **Gross Error**: These are large mistakes made by the person conducting the experiment. They result from human errors like wrong readings or miscalculations. They can be avoided with careful work.

Minimizing Errors

- **Systematic errors** can be minimized by calibrating instruments, refining experimental techniques, and accounting for known environmental effects.
- **Random errors** can be reduced by taking multiple measurements and averaging the results.
- **Gross Error:** Careful data recording, proper training, and double-checking calculations can prevent gross errors.

1.6.2. Absolute error, relative error, and percentage error

In measurements, absolute error, relative error, and percentage error are ways to express how much your measured value deviates from the true or accepted value.

1. Absolute Error

Definition: The difference between the measured value and the true (or accepted) value.

$$\text{Absolute Error} = |\text{Measured Value} - \text{True Value}|$$

Example: If the true length of an object is 10 cm and you measure it as 9.8 cm, the absolute error is:

$$|9.8 - 10| = 0.2 \text{ cm}$$

2. Relative Error

Definition: The absolute error divided by the true value, showing the error size in relation to the true value.

Relative Error = $\frac{\textbf{\textit{Absolute Error}}}{\textbf{\textit{True Value}}}$ = 0.02

Example: Using the same example as above, the relative error would be:

$\frac{\mathbf{02}}{\mathbf{10}}$ = 0.02

3. Percentage Error

Definition: The relative error expressed as a percentage, to make the error easier to interpret.

Percentage Error = $(\frac{\textbf{Absolute Error}}{\textbf{True Value}})$ *100

Example: Using the same data, the percentage error is:

Percentage Error = $(\frac{\mathbf{0.2}}{\mathbf{10}})$ *100 = 2%

In summary:

- **Absolute error** gives the actual size of the error.
- **Relative error** shows how large the error is compared to the true value.
- **Percentage error** expresses the relative error as a percentage for easier interpretation.

Exercise 1:

A student measures the length of a table and finds it to be 1.95 meters. The actual length is 2 meters. Calculate the absolute error, relative error, and percentage error.

Solution:

Measured Value: 1.95 m

True Value: 2.00 m

Absolute Error:

$$\text{Absolute Error} = |\text{Measured Value} - \text{True Value}| = |1.95 - 2.00| = 0.05 \text{ m}$$

Relative Error:

$$\text{Relative Error} = \frac{\text{Absolute Error}}{\text{True Value}} = \frac{0.05}{2.00} = 0.025$$

Percentage Error:

$$\text{Percentage Error} = \left(\frac{0.05}{2.00}\right) \times 100 = 2.5\%$$

Exercise 2:

A scale shows the mass of a bag of rice as 9.85 kg, but the actual mass is 10 kg. Find the absolute error, relative error, and percentage error.

Exercise 3:

A thermometer reads the temperature of a liquid as 75°C, but the true temperature is 80°C. Compute the absolute error, relative error, and percentage error.

1.7 Rules for Identifying Significant Digit

Significant digits (or figures) are the meaningful digits in a number, which help indicate the precision of a measurement. Here are the basic rules:

1. **Non-Zero Digits**: All non-zero digits are always significant.
   - Example: 45.67 has **4 significant digits**.
2. **Zeroes Between Non-Zero Digits**: Zeroes between non-zero digits are significant.
   - Example: 407.08 has **5 significant digits**.
3. **Leading Zeroes**: Zeroes to the left of the first non-zero digit are **not significant**; they only serve as placeholders.
   - Example: 0.00325 has **3 significant digits** (3, 2, and 5).
4. **Trailing Zeroes with a Decimal Point**: Zeroes to the right of a decimal point **are significant** if they are at the end of the number.
   - Example: 4.500 has **4 significant digits**.
5. **Trailing Zeroes without a Decimal Point**: Zeroes at the end of a number without a decimal point are **not considered significant** (unless specifically stated by context).
   - Example: 5000 has **1 significant digit** (5).
6. **Exact Numbers**: Numbers that are counted (like the number of people) or defined constants (like 1 inch = 2.54 cm) have **infinite significant figures**.

Examples of Applying the Rules of Significant Figures

Example:

1. Number: **0.00456**

- Leading zeroes are not significant.
- Significant figures: 3 (4, 5, 6)

2. Number: **6700**

- No decimal point, so trailing zeroes are not significant.
- Significant figures: 2 (6, 7)

3. Number: **102.030**

- All digits are significant because the zeroes between non-zero digits and trailing zeroes after the decimal point are significant.
- Significant figures: 6 (1, 0, 2, 0, 3, 0)

4. Number: **3.400**

- Trailing zeroes after the decimal point are significant.
- Significant figures: 4 (3, 4, 0, 0)

5. Number: **600.0**

- Trailing zeroes after a decimal point are significant.
- Significant figures: 4 (6, 0, 0, 0)

1.8. Rounding Off Numbers

When rounding a number, follow these general rules:

1. **If the first digit to be dropped is less than 5**, leave the last retained digit unchanged.
   - Example: 4.732 rounded to 2 significant figures becomes **4.7**.
2. **If the first digit to be dropped is 5 or more**, increase the last retained digit by 1.
   - Example: 4.768 rounded to 2 significant figures becomes **4.8**.
3. **If rounding to a whole number**, follow the same rule for the last non-zero digit.
   - Example: 137.6 rounded to the nearest whole number becomes **138**.

Examples of Rounding Off Numbers Using Significant Figures

1. Number: **4567**

Round to 3 significant figures

- The third significant digit is 6. The next digit is 7, which is greater than 5, so increase the last digit.
- Rounded number: **4570**

2. Number: **0.003456**

Round to 2 significant figures:

- The second significant digit is 4. The next digit is 5, so increase the last digit.
- Rounded number: **0.0035**

3. Number: **7.452**

Round to 3 significant figures:

- The third significant digit is 5. The next digit is 2, which is less than 5, so leave the last digit unchanged.
- Rounded number: **7.45**

4. Number: **845.976**

Round to 4 significant figures:

- The fourth significant digit is 9. The next digit is 7, so increase the last digit.
- Rounded number: **846.0**

1.9. Using Measuring Instruments at the Workplace

Measuring instruments are essential in various workplaces, especially in fields like engineering, construction, manufacturing, and science. They ensure accuracy, precision, and safety in measurements, which is crucial for successful operations. Here are examples of common instruments used to measure length, mass, and time.

1.9.1. Measuring Instruments Used to Measure Length

1. **Ruler/Scale**:
    - **Uses**: Measures short distances, typically in millimeters, centimeters, and inches.
    - **Application**: Useful for simple tasks like drawing straight lines or measuring small objects in manufacturing or design.
2. **Tape Measure**:
    - **Uses**: Measures longer distances, typically up to several meters.
    - **Application**: Common in construction and carpentry for measuring room dimensions, lengths of materials, etc.
3. **Vernier Caliper**:
    - **Uses**: Measures internal and external dimensions and depths with precision.
    - **Application**: Used in mechanical engineering and precision manufacturing to measure components like pipes or machine parts.
4. **Micrometre**:
    - **Uses**: Measures small objects with high accuracy, usually in the range of microns.
    - **Application**: Common in mechanical and machining industries for measuring small parts with high precision.
5. **Laser Distance Measurer**:

- **Uses**: Measures distances using a laser beam, often up to several hundred meters.
- **Application**: Used in construction, surveying, and engineering to measure room sizes, land areas, or building heights quickly and accurately.

1.9.2. Measuring Instruments Used to Measure Mass

1. **Mechanical Balance**:
    - **Uses**: Measures mass by balancing an object against known weights.
    - **Application**: Often used in educational settings, labs, or small-scale settings to weigh items with moderate precision.
2. **Electronic Balance**:
    - **Uses**: Measures mass with high precision using digital technology.
    - **Application**: Common in laboratories, pharmacies, and manufacturing industries for weighing chemicals, small parts, or pharmaceuticals.
3. **Spring Balance**:
    - **Uses**: Measures force (and indirectly mass) by the extension of a spring.
    - **Application**: Used in physics labs or certain workplaces where weight needs to be measured in relation to force (e.g., tension).
4. **Weighing Scale**:
    - **Uses**: Measures larger masses and body weight.
    - **Application**: Common in grocery stores, logistics, and hospitals to measure items or body weight.
5. **Platform Scale**:
    - **Uses**: Measures the mass of heavy or large items, typically in kilograms or tons.
    - **Application**: Used in factories, shipping docks, and warehouses to weigh heavy machinery, packages, or bulk materials.

1.9.3. Measuring Instruments Used to Measure Time

1. **Stopwatch**:
    - **Uses**: Measures precise intervals of time.
    - **Application**: Common in sports, experiments, and testing environments where accurate measurement of short time periods is necessary.
2. **Clock/Watch**:
    - **Uses**: Measures time in hours, minutes, and seconds.
    - **Application**: Everyday use to track time in offices, factories, and general workplaces.
3. **Digital Timer**:
    - **Uses**: Tracks time with precise digital display and countdown functionality.
    - **Application**: Useful in laboratories, kitchens, or manufacturing processes where activities are time-bound.
4. **Atomic Clock**:
    - **Uses**: Measures time with extremely high precision, often used to synchronize time across devices globally.
    - **Application**: Used in scientific research, telecommunications, and navigation systems.
5. **Chronometer**:

- **Uses**: High-precision timekeeping instrument, particularly used in maritime navigation.
- **Application**: Used in naval or aerospace applications where precise time measurement is critical for navigation.

LOU2. DESCRIBE MOTION IN ONE DIMENSION

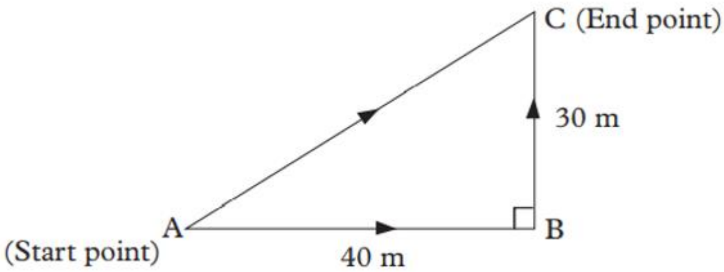
2.1. Explanation of displacement, velocity and acceleration concepts

2.1.1. Distance and Displacement

Distance is a **scalar quantity**, meaning it only has magnitude (size) and no direction.
• It refers to the total length of the path travelled by an object, regardless of the direction. It is always positive.
Example: If you walk 3 km east, then 2 km west, your total distance travelled is 5 km.

Displacement is a **vector quantity**, meaning it has both magnitude and direction.
• It refers to the shortest straight-line distance from the initial position to the final position, including direction.
• **Example**: If you walk 3 km east and then 2 km west, your displacement is 1 km east (the net change in position).
Formula for displacement: Displacement $\Delta x = x_f - x_i$
Example for calculation of distance and displacement

Suppose a boat starts at point A moves 40 km East to point B followed by 30 m North to point C as shown in Figure below.



It should be noted that the distance can be greater or equal to the displacement in magnitude.

We can determine its distance and displacement covered as follows:

- Distance = AB + BC = 40 m + 30 m = 70 m
- $Displacement = AC = \sqrt{AB^2 + BC^2} = \sqrt{40^2 + 30^2} = 50m$

2.1.2. Point Location on a Line

- To locate a point on a line, we usually use a number line or coordinate system.
- Position is described using a reference point (often x=0) and a coordinate that tells you how far along the line the point is.
- For example, if point A is located at x=5 meters, it means the point is 5 meters from the reference point.

2.1.3. Speed and Velocity

- **Speed**:
  - Speed is a **scalar quantity** that represents the rate at which an object covers distance.
  - It does not account for direction; only how fast something is moving.

$$\textbf{Speed} = \frac{\textbf{\textit{Distance}}}{\textbf{\textit{Time}}}$$

Example: If a car travels 100 km in 2 hours, its average speed is:

$$\textbf{Speed} = \frac{\textbf{\textit{100Km}}}{\textbf{\textit{2 Hour}}} = \textbf{50km/h}$$

Velocity:

- Velocity is a **vector quantity** that refers to the rate at which an object changes its position.
- It includes both the speed of the object and its direction.

$$\textbf{Velocity} = \frac{\textbf{\textit{Displacement}}}{\textbf{\textit{Time}}}$$

Example: If a car moves 100 km north in 2 hours, its velocity is

$$\textbf{Velocity} = \frac{\textbf{\textit{100 km nort}}}{\textbf{\textit{2 hours}}} = \textbf{50km/h north}$$

2.1.4. Average & Instantaneous Acceleration

- **Acceleration**:
  - Acceleration is a **vector quantity** that measures the rate of change of velocity over time. It tells you how quickly an object's velocity is changing.
  - If acceleration is positive, the object is speeding up; if negative (called deceleration), the object is slowing down.

$$\mathbf{a} = \frac{\Delta \boldsymbol{v}}{\Delta \boldsymbol{t}}$$

where $\Delta \boldsymbol{v}$ is the change in velocity, and $\Delta \boldsymbol{t}$ t is the change in time.

Average Acceleration:

- **Average acceleration** is the total change in velocity over the total time taken.

Formula: $a_{arg} = \frac{v_{f-}v_i}{t_f - t_i}$

• where $\boldsymbol{v_f}$ is the final velocity, $\boldsymbol{v_i}$ is the initial velocity, and $\boldsymbol{t_f - t_i}$ is the total time interval.

Example: If a car's velocity increases from 0 m/s to 20 m/s over 5 seconds, the average acceleration is:

$$a_{\text{avg}} = \frac{20\,\text{m/s} - 0\,\text{m/s}}{5\,\text{s}} = 4\,\text{m/s}^2$$

Instantaneous Acceleration:

- **Instantaneous acceleration** is the acceleration at a specific moment in time.
- It is found by taking the derivative of velocity with respect to time:

$$a_{inst} = \frac{dv}{dt}$$

2.2. Illustration of linear motion using corresponding graphs

Linear motion describes the movement of an object along a straight line. Its graphical representation involves the analysis of how displacement, velocity, and acceleration change with time.

2.2.1. Slopes and General Relationships

The slope represents the rate of change in a graph.

$\boldsymbol{m} = \frac{rise}{run} = \frac{\Delta_y}{\Delta_x}$ **,** where:

- Rise = vertical change (Δy)
- Run = horizontal change (Δx).

Intercept (b):

- The y-intercept of a graph is the point where the line crosses the y-axis, representing the initial value at t=0.

Equation of a Straight Line:

- The equation of a straight line is: $\boldsymbol{y = mx - b}$ where mmm is the slope and b is the intercept.

2.2.2. Graph of Displacement vs. Time

(a) Constant Velocity

- **Graph**: A straight line with a constant slope.
- **Equation:** $\boldsymbol{d(t) = v_t + d_0}$ where **v** is constant velocity an $\boldsymbol{d_0}$ is the initial displacement.
- **Interpretation**: The slope of the displacement-time graph represents the velocity. A positive slope means forward motion; a negative slope means backward motion.

(b) Constant Acceleration

- **Graph**: A parabolic curve, because the velocity is changing linearly with time.
- **Equation**: **d(t)** = $\boldsymbol{\frac{1}{2}at^2 + v_0t + d_0}$, where **a** is the constant acceleration.

- **Interpretation**: The slope of the displacement-time graph is changing, indicating acceleration.

(c) Non-Constant Acceleration

- **Graph**: The curve is irregular (non-parabolic).
- **Equation**: There is no simple form for non-constant acceleration. Displacement depends on how acceleration changes with time.
- **Interpretation**: The slope of the displacement-time graph changes irregularly, indicating a varying velocity.

2.2.3. Graph of velocity vs. Time
a. Graph with Positive Acceleration

- **Description**: Positive acceleration means that the object's velocity is increasing over time.
- **Graph**: A straight line sloping **upwards**.
- **Interpretation**: The slope of the line represents the acceleration. Since the slope is positive, the object is speeding up.

Equation: $\boldsymbol{v(t) = at + v_0}$, where **v(t)** is the velocity at time **t.**

b. Graph with Zero Acceleration

- **Description**: Zero acceleration means the object's velocity is constant (no change in speed).
- **Graph**: A **horizontal line**.
- **Interpretation**: Since the slope is zero, the object maintains a constant velocity throughout the time period.

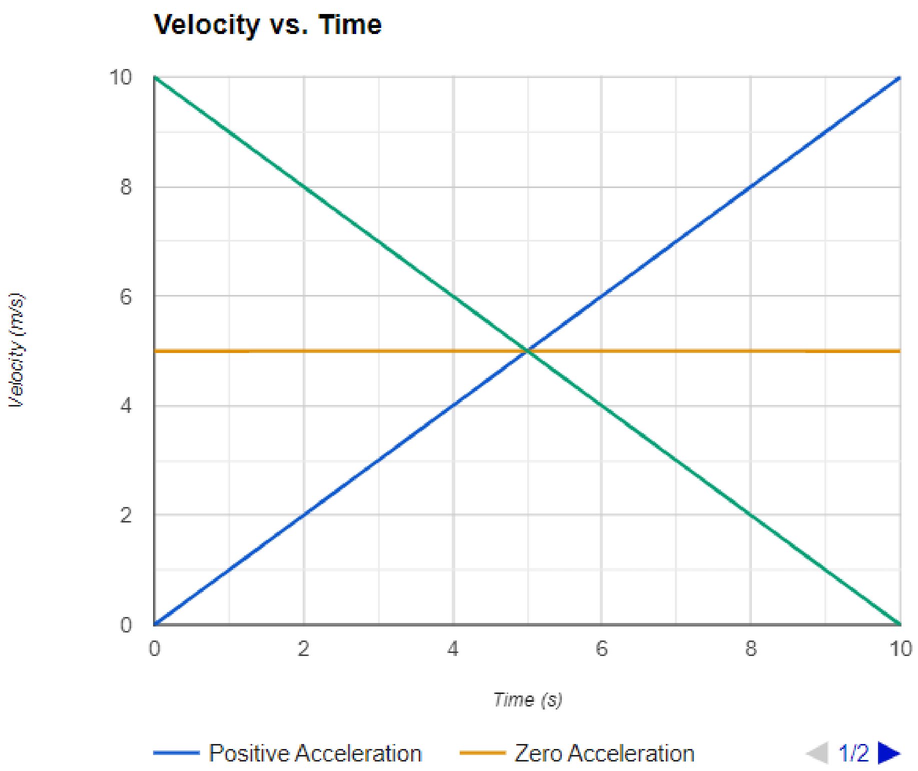
Equation: **v(t) =** $\boldsymbol{v_0}$, where $\boldsymbol{v_0}$ is a constant velocity.

c. Graph with Negative Acceleration

- **Description**: Negative acceleration (also called deceleration) means that the object's velocity is decreasing over time.
- **Graph**: A straight line sloping **downwards**.
- **Interpretation**: The negative slope indicates the object is slowing down over time.

Equation: $\boldsymbol{v(t) = -at + v_0}$**,** where:

- **a** is the magnitude of the acceleration (positive value),
- • The negative sign indicates that the velocity is decreasing.



2.3. Application of equations of motion

The equations of motion describe the relationships between displacement, velocity, acceleration, and time for objects moving with constant acceleration. These are particularly useful in analysing scenarios like free fall, where the acceleration is due to gravity.

1. Velocity as a Function of Time

The velocity of an object at any given time is directly related to its initial velocity, acceleration, and time.

$\boldsymbol{v(t) = at + v_0},$

Example:

1. An object starts from rest and accelerates at 5 m/s^2 for 8 seconds. Find its velocity after 8 seconds.

Solution

- **V_0=0 m/s (starts from rest),**
- **a=5 m/s^2**
- **t=8 second.**

$v(8) = 0 + (5)(8) = 40\,\mathrm{m/s}$, The velocity after 8 seconds is **40 m/s**.

Exercise 2: Displacement as a Function of Time
A car starts from rest and accelerates at 3 m/s^2 for 10 seconds. Find the displacement of the car after 10 seconds.

$$d(t) = v_0 t + \frac{1}{2} a t^2$$,

- **v_0 =0 m/s (starts from rest),**
- **a=3 m/s^2,**
- **t=10 seconds.**

$$d(10) = (0)(10) + \frac{1}{2}(3)(10^2) = \frac{1}{2}(3)(100) = 150\,\text{meters}$$,

The car's displacement after 10 seconds is **150 meters**.

Exercise 3: Final Velocity as a Function of Displacement

Problem:
An object is moving with an initial velocity of 10 m/s and accelerates at 2 m/s^2. Find the final velocity after the object has travelled a distance of 50 meters.

Solution

- v_0 = 10 m/s,
- a = 2 m/s^2,

- d = 50meters.

$$v^2 = v_0^2 + 2ad$$

$$v^2 = (10)^2 + 2(2)(50) = 100 + 200 = 300$$

$$v = \sqrt{300} \approx 17.32\,\text{m/s}$$

The final velocity after traveling 50 meters is approximately **17.32 m/s**.

Exercise 4: Free Fall (Velocity after Time)

Problem:
An object is dropped from a height, starting with zero initial velocity. Find the velocity of the object after 4 seconds, assuming acceleration due to gravity g=9.8 m/s^2.

Solution:
We use the velocity equation for free fall:

Where:

- v_0 =0 m/s (starts from rest),
- g =9.8 m/s^2,
- t =4seconds

$$v(4) = 0 + (9.8)(4) = 39.2\,\mathrm{m/s}\ ,$$

The velocity of the object after 4 seconds is **39.2 m/s**.

Exercise 5: Free Fall (Displacement after Time)

Problem:
An object is dropped from a certain height. Find the displacement after 6 seconds, assuming g=9.8 m/s^2.

Where:

- v_0 = 0 m/s.
- g = 9.8 m/s^2,
- t = 6 seconds.

$$d(6) = 0 + \frac{1}{2}(9.8)(6^2) = \frac{1}{2}(9.8)(36) = 176.4\,\text{meters}$$

The object has fallen **176.4 meters** after 6 seconds.

Exercise 6: Free Fall (Time to Reach Ground)

Problem:
An object is dropped from a height of 45 meters. How long will it take for the object to reach the ground? Assume g=9.8 m/s^2.

Where:

- d=45 meter,
- g=9.8 m/s^2,
- Solve for t:

$$\mathbf{d} = \frac{1}{2}\boldsymbol{g}\boldsymbol{t}^2 \ \sim\ \mathbf{45} = \frac{1}{2}(\mathbf{9.8})\boldsymbol{t}^2\ , \sim \mathbf{d} = \frac{1}{2}\mathbf{4.9}\ \boldsymbol{t}^2$$

$t^2 = \frac{45}{4.9} = 9.18$,

$t = \sqrt{9.18} \approx 3.03 \text{ seconds}$,

The object will take approximately **3.03 seconds** to reach the ground.

Graphs

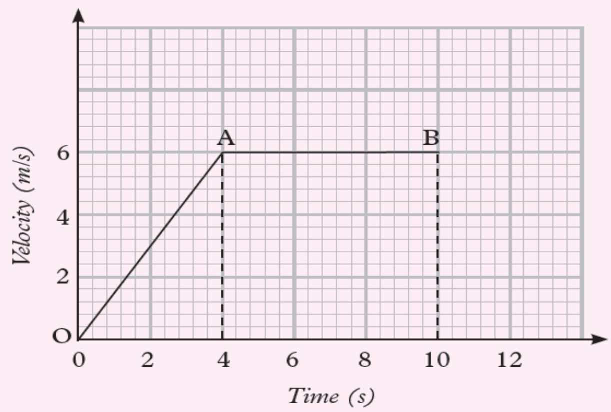
Example: Table shows the data collected to study the motion of cylist.

Velocity m/s	0	3	6	6	6	6
Time (s)	0	2	4	6	8	10

(a) Plot a graph of velocity (y-axis) against time (x-axis).
(b) Use your graph to determine the acceleration of the cyclist in the first four seconds

Solutions

(a)



(b) Acceleration = slope of the graph

$$= \frac{\text{Change in velocity}}{\text{Change in time}}$$

$$= \frac{(6-0)\text{ m/s}}{(4-0)\text{ s}} = \frac{6\text{ m/s}}{4\text{ s}}$$

$$= 1.5\text{ m/s}^2$$

LO3: ANALYE MOTION IN TWO DIMENSION

In learning outcome 2 we deal with motion along a straight line. We now consider the motion of objects that move in paths in two (or three) dimensions. Before beginning our discussion of motion in two dimensions, we will need a new tool, vectors, and how to add them.

3.1. DESCRIPTION OF SCALARS, VECTORS AND VECTOR COMPONENTS

- ❖ **Scalars**

Scalars are the quantities that are fully described by numerical value (magnitude) alone, without any reference to direction. For example, Mass, temperature, time are scalar quantities.

Properties of scalars

- Scalars are described by a single value (magnitude).
- They are represented by real numbers with units (e.g., 5 kg, 20°C).

➢ **Operation on scalars**

Addition and subtraction: Scalars can be added or subtracted using regular arithmetic, combining only their magnitudes.

Examples:

- **Temperature**: Adding two temperatures, such as combining 25°C with 5°C. The result is 25+5 = 30°C.
- **Mass**: If you have two objects with masses of 3 kg and 4 kg, their total mass 3+4=7 kg.
- **Time**: Subtracting time intervals, like taking 10 minutes and removing 3 minutes from it: 10−3=7 minutes

Multiplication and division: Scalars can be multiplied or divided by other scalars or vectors producing new scalar values.

- Examples:

- **Speed and Time** (to calculate distance): If a car is moving at 60 km/h for 2 hours, the distance travelled is calculated by multiplying the speed and time: **60 km/h×2 h =120 km.**
- **Density** (mass divided by volume): If a material has a mass of 10 kg and a volume of 2 cubic meters, its density is calculated as **10 kg÷2 m^3 =5 kg/m^3.**

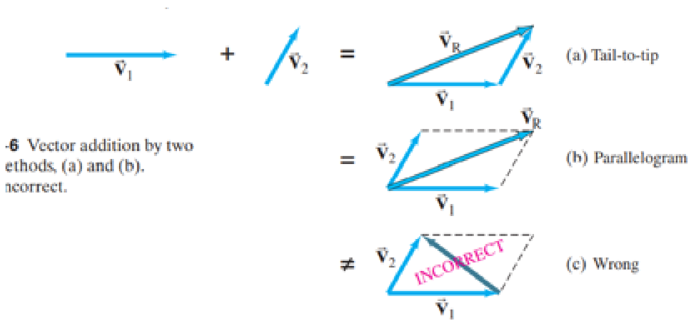
- **Work** (force and distance): If a force of 20 N is applied over a distance of 5 m, the work done is **20 N×5 m =100 J (joules).**

❖ Vectors

Vectors are fully described by both a numerical value (magnitude) and a specific direction. For example, velocity, force, and displacement are vector quantities.

Properties of Vectors

- The size or length of the vector, often represented by $|\vec{A}|$
- The orientation of the vector in space, typically given as an angle from a reference axis.
- Two vectors are equal if they have the same magnitude and direction, regardless of their position.
- The negative of a vector $\vec{A}$ is a vector with the same magnitude but opposite direction, denoted by $-\vec{A}$.
- **Zero Vector**: A vector with zero magnitude and no particular direction, represented as $\vec{0}$.



➢ Operation on vectors

Addition and subtraction:

Addition: Let $\vec{A}$ and $\vec{B}$ be two vectors. We define a new vector $\vec{C} = \vec{A} + \vec{B}$.The vector addition of $\vec{A}$ and $\vec{B}$, by geometric construction.

- Draw the arrow that represent $\vec{A}$,
- Place the tail of the arrow that represents vector $\vec{B}$ at the tips of the arrow for $\vec{A}$,
- The arrow that starts to the tail of $\vec{A}$ and goes to the tip of $\vec{B}$ is defined to be the vector addition.

- **Tip-to-Tail Method**: Place the tail of the second vector at the tip of the first vector. The resultant vector ($\vec{R}=\vec{A}+\vec{B}$) is drawn from the tail of the first vector to the tip of the second.
- **Parallelogram Method**: Place vectors $\vec{A}$ and $\vec{B}$ with the same starting point, forming a parallelogram. The diagonal of the parallelogram is the resultant vector.

Example: Imagine a person walks 3 meters east (vector $\vec{A}$) and then 4 meters north (vector $\vec{B}$). To find the total displacement:

1. Draw $\vec{A}$ (3 meters east).
2. Place the tail of $\vec{B}$ (4 meters north) at the tip of $\vec{A}$.
3. Draw the resultant vector $\vec{R}$ from the starting point (tail of $\vec{A}$) to the end point (tip of $\vec{B}$).
4. Using the Pythagorean theorem, you can find that $\vec{R}$ has a magnitude of 5 meters **(since $\sqrt{3^2+4^2}$ = 5).**

Subtraction:

- To subtract a vector $\vec{B}$ from $\vec{A}$ ($\vec{A}-\vec{B}$), add the opposite of $\vec{B}$ to $\vec{A}$ (i.e., $\vec{A}+(-\vec{B})$).
- The difference between two vectors $\vec{A}-\vec{B}$ is defined as $\vec{A}-\vec{B}=\vec{A}+\overrightarrow{(-B)}$

Note: The triangle rule of vector addition is $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$

Multiplication:

- **Scalars multiplication:** When a vector is multiplied by a scalar (a number), the result is a new vector. The new vector has a magnitude that is scaled by the scalar, while its direction remains the same if the scalar is positive, or opposite if the scalar is negative.

Example: Suppose we have a vector $\vec{A}$ representing a force of 5 N to the right. If we multiply $\vec{A}$ by a scalar of 3, **the resulting vector $3\vec{A}$ will be a force of 15 N to the right (i.e., 5×3=15 N).**

If we multiply $\vec{A}$ by −2, the result will be a vector $-2\vec{A}$ with a magnitude of 10 N in the opposite direction (to the left).

- **Dot product:** The dot product of two vectors $\vec{A}$ and $\vec{B}$ is a scalar quantity calculated by multiplying the magnitudes of the vectors and the cosine of the angle θ between them. The formula is: $\mathbf{\vec{A}\cdot\vec{B}=|\vec{A}||\vec{B}|\cos(\theta)}$

Example: Suppose $\vec{A}$ is a 10 N force directed north, and $\vec{B}$ is an 8 m displacement at a 30° angle north-east. The dot product gives the component of $\vec{A}$ in the direction of $\vec{B}$:

$\mathbf{\vec{A}\cdot\vec{B} = (10)\ (8)\ \cos(30^0) = 80\times0.866 = 69.28\ N}$

- **Cross product:** The cross product of two vectors $\vec{A}$ and $\vec{B}$ results in a new vector that is perpendicular to the plane containing $\vec{A}$ and $\vec{B}$. The magnitude of this new vector is given by:

  $\mathbf{|\vec{A}\times\vec{B}|=|\vec{A}||\vec{B}|\sin(\theta)|}$

  where **θ** is the angle between $\vec{A}$ and $\vec{B}$. The direction of the resulting vector is determined by the right-hand rule.

Example: Imagine a force vector $\vec{A}$ of 6 N applied horizontally and a displacement vector $\vec{B}$ of 4 m directed vertically (90° angle between them). The cross product would be:

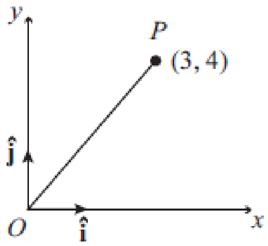
$\mathbf{|\vec{A}\times\vec{B}| = (6)(4)\ \sin(90^0) = 24\ N}$

The resulting vector would point perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$, representing a torque or rotational effect.

- **Vector components in Cartesian coordinate system**

Point and position vector

The natural way to describe the position of any point is to use Cartesian coordinates. In two dimensions, we have a diagram like this, with an x-axis and a y-axis, and an origin O. To include
vectors in this diagram, we have a vector $\vec{\imath}$ associated with the x-axis and a vector $\vec{\jmath}$ associated with the y-axis.

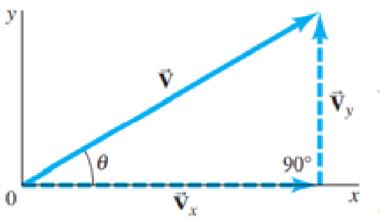


If we take any point in this diagram, for instance the point P with coordinates **(3, 4),** then we can write $\overrightarrow{\boldsymbol{OP}} = 3\vec{\imath} + 4\vec{\jmath}$**.** It is important to appreciate the difference between these two expressions. The numbers (3, 4) represent a set of coordinates, referring to the point P. But the expression $3\vec{\imath} + 4\vec{\jmath}$ is a vector, the position vector OP. An alternative way of writing this is as a "column vector":

$\binom{3}{4}$ means the same as $3\vec{\imath} + 4\vec{\jmath}$

Components of the position vector

Let consider the graph



As we see by adding V_x and V_y gives V. $V = V_x + V_y$ then V_x and V_y are **the components** of V

Using trigonometric ratios we have $\begin{cases} V_x = V\cos\theta \\ V_y = V\sin\theta \end{cases}$

The magnitude of the vector V will be given by $|\vec{V}| = \sqrt{\boldsymbol{V_x}^2 + \boldsymbol{V_y}^2}$ by Pythagorean theorem.

Examples

1. A vector V⃗ has a magnitude of 10 units and is directed at an angle of 30^0 with respect to the positive x-axis. Find the components of the vector V⃗ along the x- and y-axes.

Solution

- **V = 10units**
- **Θ = 30^0**

$$V_x = 10\cos(30^\circ) = 10 \times 0.866 = 8.66 \text{ units}$$

$\begin{cases} V_x = V\cos\theta \\ V_y = V\sin\theta \end{cases} \cong$ $$V_y = 10\sin(30^\circ) = 10 \times 0.5 = 5 \text{ units}$$

2. Given the components of a vector $\vec{V}$ as $V_x = 8$ units and $V_y = 6$ units, find the magnitude of the vector $\vec{V}$.

$$|\vec{V}| = \sqrt{V_x^2 + V_y^2} \quad |\vec{V}| = \sqrt{(8)^2 + (6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ units}$$

3. A vector $\vec{V}$ has components $V_x = 7$ units and $V_y = 24$ units. Find the angle θ of the vector with respect to the positive x-axis.

Solution

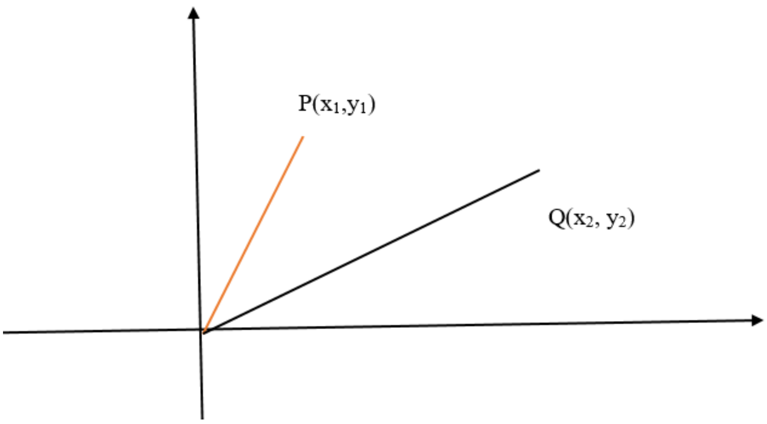
We can find the angle θ using the following trigonometric ratio:

$$\tan\theta = \frac{V_y}{V_x}, \quad \tan\theta = \frac{24}{7}, \quad \theta = \tan^{-1}\left(\frac{24}{7}\right) = \tan^{-1}(3.43) \approx 74.74^\circ$$

3.2. ILLUSTRATION OF DISPLACEMENT, VELOCITY AND ACCELERATION

- **Displacements in two dimensions**

Suppose a particle moves on plane along the curve as shown in the figure below. At time t_1 the particle is at the point P and some later time t_2, the particle is at Q. As the particle moves from P to Q in the time interval $\Delta t = t_2 - t_1$, the position vector changes from $r_1 = OP$ to $r_2 = OQ$.



From triangle rule of vector addition, we can see that:

$\overrightarrow{OP} + \overrightarrow{PQ} = \overrightarrow{OQ}$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = r_2 - r_1 = \Delta r$

Here $r_1 = x_1\vec{i} + y_1\vec{j}$ and $r_2 = x_2\vec{i} + y_2\vec{j}$

$\Delta r = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} = \Delta x\vec{i} + \Delta y\vec{j}$

- **Velocity vector**

The Average Velocity Vector

We define average velocity during time interval as the ratio of the displacement of that interval time. $\overrightarrow{V_{av}} = \frac{\overrightarrow{\Delta r}}{\Delta t} = \frac{\Delta x}{\Delta t}\vec{i} + \frac{\Delta y}{\Delta t}\vec{j} = V_x + V_y$

The Instantaneous Velocity Vectors

Instantaneous velocity vector: is the limit of the average velocity as Δt approaches zero. Its direction is along a line that is tangent to the path of the particle and in the direction of motion.

$$\overrightarrow{V_{in}} = \lim_{\Delta t \to 0} \frac{\Delta r}{\Delta t}$$

The magnitude of the velocity is given by: $|\vec{V}| = \sqrt{V_x{}^2 + V_y{}^2}$ and $\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right)$

Example1: A sailboat has coordinates (130 m, 205 m) at t1=0.0 s. Two minutes later its position is (110 m, 218 m). Find its average velocity and magnitude of the average velocity.

Solution

- **Initial position at t_1 = 0.0 s: (x_1, y_1) = (130 m, 205 m)**
- **Final position at t_2 =120 s: (x_2, y_2) = (110 m, 218 m)**

Note: The time interval is given as two minutes, which is 120120120 seconds.

$$\Delta \vec{r} = (x_2 - x_1, y_2 - y_1)$$

$$\Delta x = x_2 - x_1 = 110\,\mathrm{m} - 130\,\mathrm{m} = -20\,\mathrm{m}$$

Plugging in the values: $\Delta y = y_2 - y_1 = 218\,\mathrm{m} - 205\,\mathrm{m} = 13\,\mathrm{m}$

$$\Delta \vec{r} = (-20\,\mathrm{m}, 13\,\mathrm{m})$$

Average Velocity Vector $\vec{v}_{\mathrm{avg}}$:

$$v_{\mathrm{avg},x} = \frac{-20\,\mathrm{m}}{120\,\mathrm{s}} = -0.167\,\mathrm{m/s}$$

$$\vec{v}_{\mathrm{avg}} = \frac{\Delta \vec{r}}{\Delta t} = \left(\frac{\Delta x}{\Delta t}, \frac{\Delta y}{\Delta t}\right) \qquad v_{\mathrm{avg},y} = \frac{13\,\mathrm{m}}{120\,\mathrm{s}} = 0.108\,\mathrm{m/s}$$

So, the average velocity vector is:

$$\vec{v}_{\mathrm{avg}} = (-0.167\,\mathrm{m/s}, 0.108\,\mathrm{m/s})$$

Calculate the Magnitude of the Average Velocity: To find the magnitude of $\vec{v}_{avg}$, we use the Pythagorean theorem:

$$|\vec{v}_{\text{avg}}| = \sqrt{v_{\text{avg},x}^2 + v_{\text{avg},y}^2}$$

Substituting the values:

$$|\vec{v}_{\text{avg}}| = \sqrt{(-0.167\,\text{m/s})^2 + (0.108\,\text{m/s})^2}$$

$$|\vec{v}_{\text{avg}}| = \sqrt{0.0279 + 0.0117}$$

$$|\vec{v}_{\text{avg}}| = \sqrt{0.0396}$$

$$|\vec{v}_{\text{avg}}| \approx 0.199\,\text{m/s}$$

Example2: A dragonfly is observed initially at position: $\vec{r}_1 = (2.00m)\vec{\imath} + (3.50m)\vec{\jmath}$. Three seconds later, it is observed at position: $\vec{r}_2 = (-3.00m)\vec{\imath} + (5.50m)\vec{\jmath}$. What was the dragonfly's average velocity during this time?

- ❖ **Acceleration Vectors**

The average acceleration vector is defined as the rate at which the velocity changes. It is in the direction of the change in velocity.

$$\vec{a}_{av} = \frac{\Delta\vec{V}}{\Delta t}$$

The instantaneous acceleration is the limit of the average acceleration as Δt approaches zero.

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t} = a_x + a_y$$

The velocity vector $\vec{V}$ always points in the direction of motion. The acceleration vector $\vec{a}$ can point anywhere.

Example: A Thrown Baseball The position of a thrown baseball is given by: $\vec{r} = \left[1.5m + \left(\frac{12m}{s}\right)t\right]\vec{\imath} - [\left(\frac{16m}{s}\right)t - \left(\frac{4.9m}{s2}\right)t^2]\vec{y}$. Find the velocity and the acceleration as a function of time.

3.3 ANALYZE MOTION IN TWO DIMENSIONS

- ❖ **Motion in two dimension**

The motion is said to be two dimensional, if two or three coordinates are required to specify the position of the object in space changes with respect to time.

- ❖ **Projectile motion**

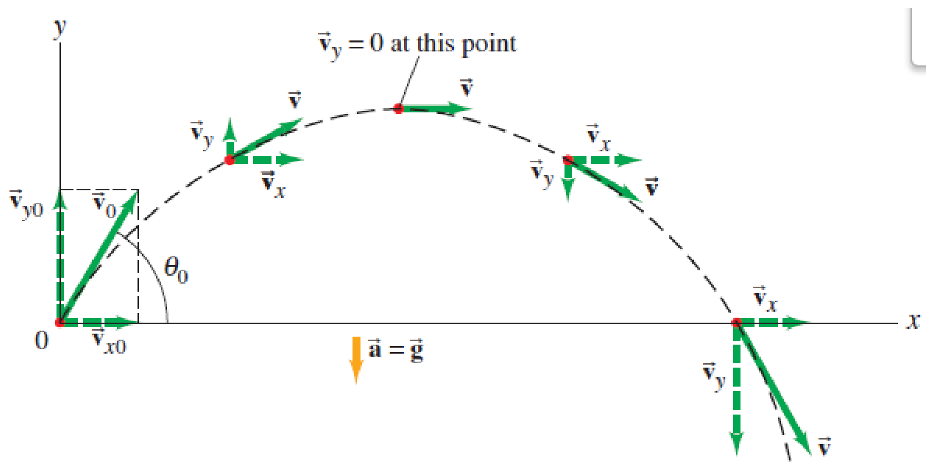
We can define **a projectile as anybody thrown into space/air**. The path taken is called **a trajectory**. The motion of a projectile unless taken otherwise is a free motion under gravity. We assume that **air resistance is negligible** in this kind of motion.

Application of projection motion

- ✓ Football player kicking a ball at a certain angle if the ball is kicked so that it does not roll on the ground, it will move at certain angle relative to the ground.
- ✓ The launch of missile in modern warfare.
- ✓ A fireman wishes to know the maximum height on the wall he can project water from the hose.

Projection angle: It is the angle formed by the initial velocity of the body and the horizontal axis through which the body is thrown.

- ❖ **Equation of motion**



This is the motion in the x-y plane; we consider axis OX and OY. $\vec{v_o}$ has two components even the acceleration. We have:

For the acceleration: $\begin{cases} \overrightarrow{a_x} = 0 \\ \vec{a}_y = -\vec{g} \end{cases}$

For Velocity: $\begin{cases} V_x = u_x - at = ucos\,\theta \\ \\ V_y = u_y + a_yt = usin\,\theta - gt \end{cases}$

According to OX axis, we have the rectilinear uniform motion whose velocity is constant and has value $v_{ox} = v_o \cos\alpha$= constant

According to OY axis, we have the rectilinear uniformly decelerated motion with acceleration –gt.

Horizontal motion

$\overrightarrow{a_x} = 0$, $u_x = ucos\,\theta$ and $x = u_xt + \frac{1}{2}a_xt^2$, we have $\boldsymbol{x = ut\cos\theta} \rightarrow (\mathbf{1})$

Vertical motion

$\boldsymbol{V_y = u_y + a_yt = usin\,\theta - gt} \;\; \rightarrow \;\; (\mathbf{2})$

$$\boldsymbol{Height\; y = u_yt + \frac{1}{2}a_yt^2 = ut\;\sin\theta - \frac{1}{2}gt^2} \quad \rightarrow (\mathbf{3})$$

Equations (1) and (3) represent the parametric equations of the motion. Using the equations developed above, obtain the parametric equation. We have:

$$\boldsymbol{y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{u^2cos^2\theta}\right)} \quad \rightarrow (\mathbf{4})$$

Maximum height

Let y_{max} be the maximum height reached by projectile. $y = y_{max}$ if and only if $v_y = 0$ (means $\boldsymbol{0 = usin\,\theta - gt}$).

$$\boldsymbol{t = \frac{usin\,\theta}{g}}$$

$\rightarrow (\mathbf{5})$ ***This the time used by projectile to reach the maximum height.***

When we consider equation 5 and 3, we see that the maximum height is given by:

$$\boldsymbol{h_{max} = \frac{u^2sin^2\theta}{2g}}$$

The Horizontal Range of the Projectile

The horizontal distance travelled by a projectile from the initial position ($x = y = 0$) to the position where it passes y = 0 during its fall is called **the horizontal range R.** Horizontal rage also is the distance travelled during the time of flight. To calculate this range denoted R, it's important to know that $R = x_{max}$ when $y = 0$

By solving the equation 4 $\boldsymbol{y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{u^2\cos^2\theta}\right) = 0}$ we have $x = 0$ $(At\ initial\ position)$ or $\boldsymbol{R = x_{max} = \frac{u^2\sin 2\theta}{g}}$

Note: the total time of flight is double of the time of reaching maximum height, i.e $t = 2\frac{\boldsymbol{u\sin\theta}}{\boldsymbol{g}}$

<u>Exercises about projectile motion</u>

1. A football is kicked with an initial velocity u = 20 m/s at an angle θ=30^0 with the horizontal. Calculate the **maximum height** reached by the projectile.

Solution

Where: $\boldsymbol{u}$ = 20 m/s

θ = 30^0

g = 9.8 m/s^2

$\sin 30^0 = 0.5$

$$\boldsymbol{h_{max} = \frac{u^2\sin^2\theta}{2g}}$$

$$h_{\max} = \frac{(20)^2(0.5)^2}{2(9.8)}$$

$$h_{\max} = \frac{400 \times 0.25}{19.6}$$

$$h_{\max} = \frac{100}{19.6} \approx 5.1\,\mathrm{m}$$

2. A projectile is launched with an initial velocity u = 25 m/s at an angle θ = 45^0 Calculate the total time of flight**.**

Solution

Where: $\boldsymbol{u}$ = 25 m/s

θ = 45^0

g = 9.8 m/s^2

$\sin 45^0 = \frac{\sqrt{2}}{2} = 0.707$, $\boldsymbol{t = \frac{2\,u\sin\theta}{g}}$

$$\boldsymbol{t = \frac{2\,u\sin\theta}{g} = \frac{2\,(25)\,\sin 45}{9.8} = 2\,\frac{(25)\,0.707}{9.8} = \frac{42.54}{9.8} = 4.34\ \mathrm{s}}$$

3. A projectile is launched with an initial velocity u = 30 m/s at an angle θ = 60^0. Calculate the horizontal range of the projectile.

Solution

Where: $\boldsymbol{u}$ = 30 m/s

θ = 60^0

g = 9.8 m/s²

$$\boldsymbol{R = x_{max} = \frac{u^2 \sin 2\theta}{g}}$$

First, calculate $\sin 2\theta$. Since $\theta = 60^\circ$, $2\theta = 120^\circ$:

$$\sin 120^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.866$$

$$R = \frac{(30)^2(0.866)}{9.8}$$

$$R = \frac{900 \times 0.866}{9.8}$$

$$R \approx \frac{779.4}{9.8} \approx 79.5\,\mathrm{m}$$

5. A ball is thrown with an initial velocity u=15 m/s at an angle θ = 45^0. Find the **height** and horizontal position of the ball at t = 1.

Solution

Where: $\boldsymbol{u}$ = 15 m/s

θ = 45^0

g = 9.8 m/s²

t = 1 s

(a) Height (y): $y = u \sin\theta\, t - \frac{1}{2}gt^2$

First, calculate $\sin 45^\circ = 0.707$. Substitute values:

$$y = (15)(0.707)(1) - \frac{1}{2}(9.8)(1)^2$$

$$y = 10.605 - 4.9 = 5.705\,\mathrm{m}$$

(b) Horizontal Position (x): $x = u\cos\theta\, t$

Where $\cos 45^\circ = 0.707$. Substitute values:

$$x = (15)(0.707)(1)$$

$$x = 10.605\,\mathrm{m}$$

LO4: DEMONSTRATE ELECTROSTATIC PHENOMENA

IC4.1: DESCRIPTION OF ELECTROSTATIC CHARGES AND THEIR CONSERVATION

Electric charges

- **Electrostatics**: It is a branch of electricity that deals with the study of electric charges at rest
- The structure of atoms can be described in terms of three particles: the negatively charged electron, the positively charged proton, and the uncharged neutron.
- **The elementary charge**, usually denoted by *e* is the electric charge carried by a single proton or, equivalently, the magnitude of the negative electric charge carried by a single electron, which has charge $-1\ e$. This elementary charge is a fundamental physical constant.
- The protons and neutrons in an atom are located in the nucleus. The charge of electron is (e-) = -1.6×10^{-19} C that of proton is (p+) = $+1.6\times10^{-19}$ C. Neutron is neutral particle, it has no charge.
- There are two types or kinds of electric charges in nature. **Negative** charge and **Positive** charge.

Laws of electrostatic charges

The law states that the like charges (charges of the same sign) repel one another and unlike charges (charges with opposite signs) attract one another.

Conservation of electric charges

The Law of conservation of electric charges states that the net amount of electric charge produced in any process is zero.

Electrification (charging) methods

- Charging by rubbing
- Charging by contact
- Charging by induction

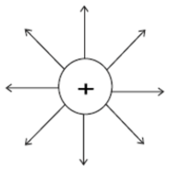
Electrostatic field

- **An electric field** is a space around the charged object where force is exerted on the charged particle.
- **Electric field lines** (lines of force) in an electric field are imaginary lines drawn through an area or place.

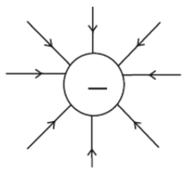
The rules for drawing electric field lines

- Electric field lines therefore point away from positive charges and towards negative charges.
- Field lines are drawn closer together where the field is stronger.
- Field lines never cross.

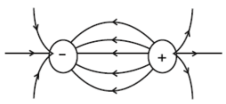
▪ Electric field lines produced by a single positive point charge.



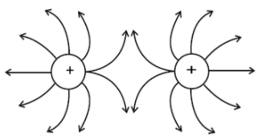
▪ Electric field lines produced by a single negative point charge.



▪ Electric field lines produced by two equal and opposite point charges.



▪ Electric field lines for two equal and positive charges.



> Note: The number of lines leaving the positive charge equals the number entering the negative charge.

Electric field at a point charge

- An electric field is also described as the electric force per unit charge.
- The formula of electric field is given as E = F/Q.

Example

1. **A** charge of Q = 2 C experiences a force of F = 10 N in an electric field. What is the strength of the electric field?

$$\mathbf{E} = \frac{F}{Q} = \frac{10\ N}{2\ C} = \mathbf{5\ N/C}$$

2. An electric field of strength E = 20 N/C exerts a force of F=4 N on a charge. What is the magnitude of the charge?

$$Q = \frac{F}{E} = \frac{4\ N}{20\ N/C} = 0.2\ C$$

Electric field between two charges

The electric field (E) between two charges is a vector field that describes the force experienced by a positive test charge placed in that region. The electric field is generated by the charges and provides information about the force that a positive charge would experience at any point in the field.

IC4.2: DETERMINATION OF THE ELECTROSTATIC FIELD

Coulomb's law of electrostatic charges

The law states that "Two electrically charged bodies experience an attractive or repulsive force F, which is inversely proportional to the square of the distance(r) between them and directly proportional to the product of their electric charges Q1 and Q2", that is:

$F \propto \frac{Q_1 Q_2}{r^2}$ or $F = k\frac{q_1 q_2}{r^2}$ whereby k= coulomb's constant $\left(k = \frac{1}{4\pi\varepsilon_0} \right)$ and ε_0 is permittivity in free space = **8.85 x10^{-12} c^2 / Nm^2**

$k = 9\text{x}10^9\ Nm^2/c^2$

therefore $\boxed{F = k\frac{q_1 q_2}{r^2}}$

Force is in newtons (N), the charges in coulombs(C) and distance (d) in metres (m).

Coulomb (C) can also be expressed as:

1 coulomb = 10^{-6} micro coulombs

1 coulomb = 10^{-9} nano coulombs

Example:

1. Suppose two-point charges each with a charge of +1.0 C are separated by a distance of 1m. Determine the magnitude of the electrostatic force between them. Is the force attractive or repulsive? (k = 9 × 109 Nm2c^{-2}).

Data given

$Q_1 = +1.0\ C, Q_2 = +1.0\ C,\ d = 1\ m$

Since, $F = k\frac{Q_1 . Q_2}{d^2} = \frac{9 \times 10^9 \times 1.0 \times 1.0}{1^2}$ then,

$F = 9 \times 10^9\ N$. The force is repulsive since it is from two similar charges.

2. Two-point charges, q1=5 μC and q2=−3 μC, are separated by a distance of r = 0.4m. Calculate the force between the charges.

Solution

- K = 9×109 N (Coulomb's constant)
- q_1 =5 μC = 5×10^{-6} C
- q_2 = −3 μC = −3×10^{-6} C
- r = 0.4 m

$$F = (9 \times 10^9)\frac{|(5 \times 10^{-6})(-3 \times 10^{-6})|}{(0.4)^2}$$

$$F = (9 \times 10^9)\frac{15 \times 10^{-12}}{0.16}$$

$$F = (9 \times 10^9) \times 93.75 \times 10^{-12}$$

$$F = 843.75 \times 10^{-3} = 0.84375\,\mathrm{N}$$

3. Two charges q_1 = 2 μC and q_2 = 8 μC exert a repulsive force of F = 0.5 N. Find the distance r between the charges.

- K = 9×109 N (Coulomb's constant)
- q_1 =2 μC = 2×10^{-6} C
- q_2 = 8 μC = 8×10^{-6} C
- F = 0.5 N

$$r = \sqrt{\frac{(9 \times 10^9)(2 \times 10^{-6})(8 \times 10^{-6})}{0.5}}$$

$$r = \sqrt{\frac{(9 \times 10^9)(16 \times 10^{-12})}{0.5}}$$

$$r = \sqrt{\frac{144 \times 10^{-3}}{0.5}}$$

$$r = \sqrt{288 \times 10^{-3}} = \sqrt{0.288} \approx 0.537\,\mathrm{m}$$

4. Two charges q_1 = 6 μC and q2=? q_2 =? q_2 =? are separated by r = 0.5m. If the force between them is F=1 N.

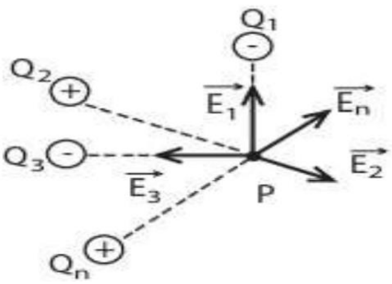
Electric field intensity and potential

The intensity of an electric field at any point is determined by the force acting on a unit positive charge (+C) placed at that point. If a positive point charge **Qo** (test charge) is placed at any point in an electric field and it experiences an electric force **F**, Electric field (E) is the electric force (F) divided by the magnitude of the test charge (Q).

- The force acting on a charge in an electric field is given by $F = QE$
- Its unit is newton per coulomb **N/C**
- The electric field is a vector quantity.
- If charge is negative, then the electric field is will be negative otherwise it is positive

Electric field strength due to distribution of electric field

Superposition is the term used to describe the addition of two or more electric fields to yield or to produce another. Consider many point charges Q1, Q2, Q3…Qn and their corresponding electric fields caused by the individual point charges E1, E2, E3 … En. The resultant electric field (**net Electric field**) at a point **p** is **the vector sum of the field** at p due to each point charge distribution.



E = E1 + E2 + E3 + ….+ En

Electric field potential and its mathematical treatment

The electric potential V (at a point P) is defined as the work done (W) in moving the positive test charge q from a large distance (infinity) to that point.

$$W = F \cdot d,\quad W = \frac{QQ_o}{4\pi\varepsilon d^2} \cdot d,\ \text{then}\ W = \frac{QQ_o}{4\pi\varepsilon\, d}$$

Then $W_{\infty - A} = U_f - U_i$

- Potential energy of test charge Q_o at d= ∞

Is given by $U_\infty = \dfrac{QQ_o}{4\pi\varepsilon\, d_\infty}$

Example:

1. A charge Q = 5μC creates an electric field. A test charge Q_0 =2μC is moved to a distance of d = 0.1 m from the source charge. Calculate the work done. Assume $\epsilon_0 = 8.85\times10^{-12}$ F/m (permittivity of free space).

Solution

$$W = \frac{QQ_0}{4\pi\epsilon_0 d},$$

$$W = \frac{(5\times10^{-6})(2\times10^{-6})}{4\pi(8.85\times10^{-12})(0.1)}$$

$$W = \frac{10\times10^{-12}}{1.11\times10^{-11}}$$

$$W \approx 0.9\,J$$

2. A charge Q = 10μC is fixed in space. A test charge Q_0 = 4μC is placed at a distance d = 0.2m. Calculate the potential energy of the system.

Work Done in Moving a Charge Between Two Points

By definition , $V = \frac{W}{Q}$, **$V2 - V1 = \frac{W}{Q}$,**

$W = Q(V_2 - V_1)$ Is work done for bringing a positive charge Q from B to A

Examples

1. A charge of Q = 2μC is moved from a point with potential V_1 =20V to another point with potential V_2 = 50 V. Calculate the work done.

Solution

$$W = Q(V_2 - V_1)$$

$$W = (2\times10^{-6})(50-20)$$

$$W = 2\times10^{-6}\times30 = 6\times10^{-5}\,\text{J},$$ Answer: $W = 6\,\mu J$.

2. A charge of Q = 5nC is moved between two points with potentials $V_1 = -10$V and $V_2 = 25$V. Find the work done.

Solution

$$W = Q(V_2 - V_1)$$

$$W = (5\times10^{-9})(25-(-10))$$

$$W = 5\times10^{-9}\times35 = 1.75\times10^{-7}\,\text{J},$$ Answer: $W = 0.175\,\mu J$.

Electric Potential and Potential Energy

$\mathbf{V} = \frac{U}{Q_o}$, then $\mathbf{U = VQ_o}$, $\frac{QQ_o}{4\pi\varepsilon\, d} = \mathbf{VQ_o}$

Then $\mathbf{V} = \frac{Q}{4\pi\varepsilon\, d}$

Example:

1. A test charge $Q_0 = 3\mu C$ is placed at a point where the electric potential is $V = 40V$. Calculate the potential energy of the test charge.

Solution

$$U = (40)(3 \times 10^{-6})$$

$U = VQ_0 \quad U = 1.2 \times 10^{-4}\,\mathrm{J}$ **Answer:** $U = 120\,\mu J.$

2. A charge Q = 8nC creates an electric field at a point d = 2m. Find the electric potential at this point. Use $\epsilon = 8.85\times10^{-12}$ C2/N.

Solution

$$V = \frac{8 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(2)}$$

$$V = \frac{8 \times 10^{-9}}{222.32 \times 10^{-12}}$$

$V = \frac{Q}{4\pi\epsilon d}$, $\quad V = 35.96\,\mathrm{V}$

Effect of an electrostatic field on a moving charge

- When a charged particle moves through an electrostatic field, it experiences a force due to the interaction between its charge and the electric field. This force can lead to both **deflection and acceleration** of the charged particle.
- The force (FF) experienced by a charged particle of charge (q) moving with velocity (v) in an electric field (E) is given by the equation: F = q·E. This force can be decomposed into two components: one parallel to the velocity vector (v) and one perpendicular to it. The component parallel to the velocity affects the particle's acceleration (a), and the component perpendicular to the **velocity causes deflection.**

IC4.3: DEMONSTRATION OF EFFECTS OF ELECTRIC FIELD ON CHARGED PARTICLES

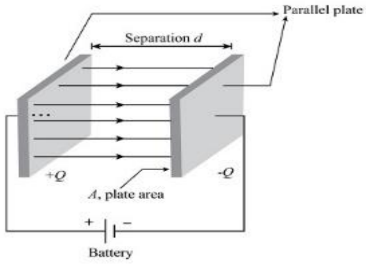
Capacitor

Capacitor is an electronic component capable of storing electrical energy in electric field, especially one consisting of two conductors separated by dielectric.

	Capacitor (C)	It acts as short circuit with a.c and open with d.c

Capacitance (C) is a fundamental property of a capacitor that quantifies its ability to store electric charge per unit voltage. It is defined as the ratio of the magnitude of the electric charge (Q) stored on one plate of the capacitor to the voltage (V) across the plates. Mathematically, capacitance is expressed by the formula: $\boldsymbol{C = \frac{Q}{V}}$**.** Where Q is electric charge, V is potential difference. Its SI unit is **Faraday.**

Parallel Plate Capacitors are formed by an arrangement of electrodes and insulating material or dielectric. A parallel plate capacitor can only store a finite amount of energy before dielectric breakdown occurs.

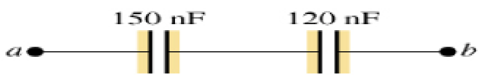


Effective capacitance for capacitor network

- **In series**

Capacitors can be connected in series: The equivalent capacitance for series-connected capacitors can be calculated as

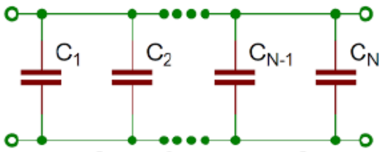
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$



- **In parallel**

When simplified, is the expression for the equivalent capacitance of the parallel network of three capacitors:

$$C_n = C_1 + C_2 + \cdots + C_n$$



$$C_{Tot} = C_1 + C_2 + \ldots + C_{N-1} + C_N$$

Electrostatic energy stored by a capacitor

The energy stored in a capacitor is electrostatic potential energy and is thus related to the charge Q and voltage V between the capacitor plates. A charged capacitor stores energy in the electrical field between its plates. As the capacitor is being charged, the electrical field builds up. When a charged capacitor is disconnected from a battery, its energy remains in the field in the space between its plates. $E = \frac{1}{2}cv^2$

Examples of electrostatic phenomena

- **Electrostatic discharge (ESD)**: It is the release of static electricity when two objects come into contact. **E.g.** A balloon rubbed against one's hair.

- **Lightning arrestors**: Arrestors are typically installed near critical appliances or points of entry, such as an electrical panel or near a generator. When potentially dangerous lightning strikes, the arrestor activates and diverts the lightning to ground where it will disperse harmlessly

- **Paint spraying**: A positively charged electron within the spray nozzle charges the paint particles. Because these particles all have a positive charge, they repel each other and break apart, resulting in a fine mist coat evenly.

- **Photocopier machines**: Negatively charged powder spread over the surface adheres through electrostatic attraction to the positively charged image areas.

LO5: APPLY GEOMETRIC OPTICS

5.1. DESCRIBE PROPERTIES OF LIGHT IN HOMOGENEOUS MEDIUM

- **Key terms:**

Optics: is the study of the behaviour and physical properties of light. This has led to development of various optical devices like the lenses that are used in cameras by people with eye defects, projectors, microscopes, telescope, fibre optics among others.

Geometrical optics: is **a model of optics that describes light propagation in terms of rays**. The ray in geometric optics is an abstraction useful for approximating the paths along which light propagates under certain circumstances.

Light: the natural agent that stimulates sight and makes things visible.

- **Natural light:** the light from the sun**.** Natural lighting, also known as **daylighting**, is a technique that efficiently brings natural light into your home using exterior glazing (windows, skylights, etc.)
- **Source light:** is anything that makes light, whether natural and artificial. Natural light sources include the Sun and stars.

Format principle

Fermat's principle, in optics, statement that light traveling between two points seeks a path such that the number of waves (the optical length between the points) is equal, in the first approximation, to that in neighbouring paths.

5.1.1 Sources of light

(a) Luminous sources of light

These are sources (objects) that emit (give out) their own light.

Examples of non-living luminous objects are **sun, stars**, **fire**, candle flame and electric bulb.

Examples of living things that are luminous objects are **fireflies and glow worm**.

(b) Non-luminous sources of light

These are objects that do not emit (give out) their own light. We get to see these objects when they reflect the light falling on them from luminous source onto our eyes.

The moon is a good example of a non-living thing that is non-luminous source of light. Others are a wall and a car.

Examples of a living things that are non-luminous sources are trees and animals.

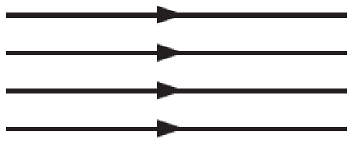
5.1.2 Propagation of light

Rays and beams

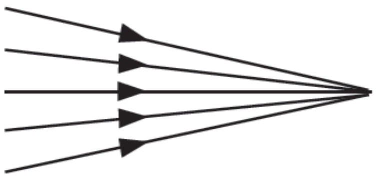
A ray *of light is the path along which light travels in a medium.*

A beam *of light is a collection or group of light rays. There are three types of beam of light rays.*

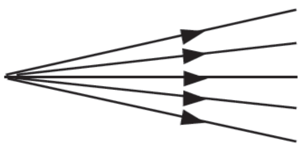
(a) Parallel beam: consists of rays that are parallel to one another



(b) **Convergent beam:** consists of rays of light that meet at a point



(c) **Divergent beam:** consists of rays of light originating from a point source and diverge(spread) to different directions.



The properties of light

- ✓ Light is a form of energy.
- ✓ It enables us to see the surrounding objects.
- ✓ Light itself is not visible but its effect is felt by the eye.
- ✓ It travels at a speed of approximately 300 000 000 m/s or 3×10^8 m/s.
- ✓ It always propagates on straight line. (Formation of shadow and eclipse)
- ✓ Light does not need any medium for its propagation (Light can travel in vacuum)

Medium of propagation

Transparent, translucent and opaque materials

Transparent materials – These are materials that allow all the light falling on them to pass through them freely. Therefore, we are able to see clearly through these materials.

Examples of transparent materials are air, water and clear glass.

Translucent materials – These are materials that allow some light falling on them to pass through. The light gets scattered as it passes through. Therefore, objects on the other side of such materials appear blurred and cannot be seen clearly.

Examples of translucent materials are frosted glass, oiled paper, wax paper, ice, tinted windows and some plastics.

Opaque materials – These are materials that do not allow light to pass through. When light strikes an opaque object, none of it passes through. Therefore, we cannot see through such materials. When light falls on these materials, much of it is reflected away by the objects some while of it absorbed and converted to heat energy.

Examples of opaque materials are rocks, wood, soil, metals and exercise book.

5.2 APPLICATION OF LIGHT REFLECTION ON SURFACES

Reflection is the throwing back by a body bouncing back off of light as it strikes a surface. The ray coming from the source is called **incident ray**. The ray moving away from the reflecting surface is called **reflected ray**.

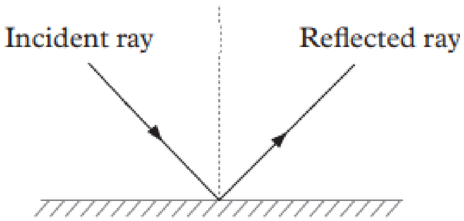


Fig 13.17: A reflection of light

Types of reflection

There are two types of reflection; **regular and diffuse** (irregular) reflections.

- ✓ When light is reflected by a plane or a smooth surface, the **reflection is regular** i.e parallel incident rays are reflected parallel to each other.
- ✓ When reflection occurs at a rough surface, it is called **diffuse reflection** i.e incident parallel rays are reflected in random directions. Figures below show the two types of reflections.

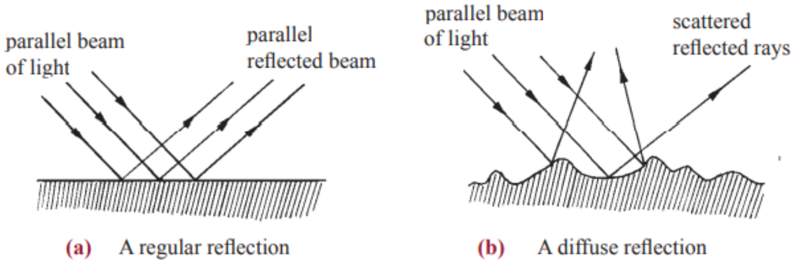


Fig. 13.18 : Reflection of light on different surfaces

Laws of reflection

The laws of reflection of light state that:

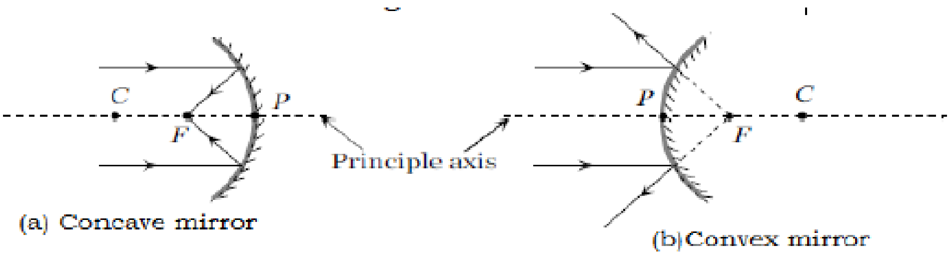
1. The incident ray, reflected ray, and normal at the point of incidence all lie in the same plane.
2. The angle of incidence is equal to the angle of reflection.

Formation of image in Spherical or curved mirrors

A **spherical mirror** is a mirror that has the reflecting surface which is curved. There are two types

Concave Mirrors: The reflecting surface is curved **inwards**.

Convex Mirrors: The reflecting surface is curved **outwards**.



Magnification

Let u =object distance
v =image distance
f =focal length
R =radius of the curvature
The magnification

$$M = \frac{h'}{h} = \frac{v}{u}$$

Magnification has no unit.

The distance v can be found from the expression $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Since $R = 2f$, we also have $f = \frac{R}{2}$

Note: In calculation, f and v are taken as ***positive*** for concave mirror whereas for convex mirror they are taken as ***negative***.

Applications of spherical/curved mirrors

i. Concave mirror: Used as a shaving mirror, In search light, in cinema projector, in telescope, etc.

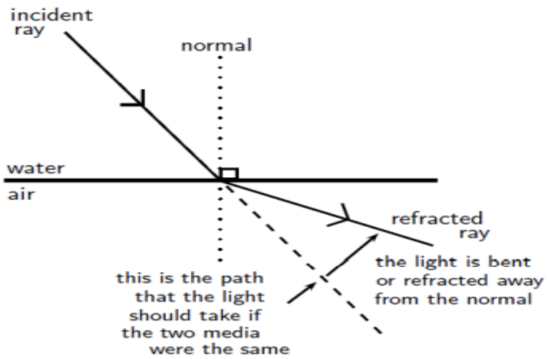
ii. Convex mirror: In road lamps, side mirror in vehicles etc.

5.3 APPLICATION OF LIGHT REFRACTION IN DIFFERENT MEDIA

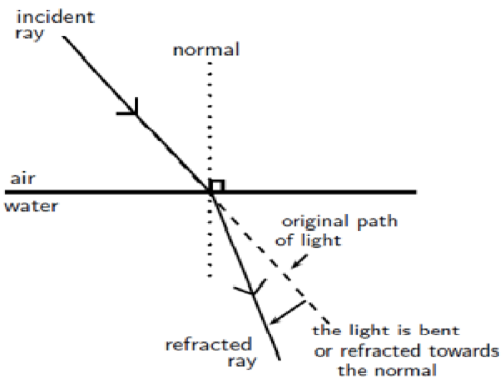
Refraction is the bending of light that occurs because light travels at different speeds in different materials.

Interface is a boundary between two regions of space occupied by different matter in different physical states

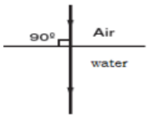
Light is moving from an optically dense medium to an optically less dense medium. Light is refracted away from the normal.



Light is moving from an optically less dense medium to an optically denser medium. Light is refracted towards the normal.



When an incident light is perpendicular to the surface, it continues to travel undeviated in a straight line but the speed of light is reduced in the glass. This is, sometimes, referred to as the **normal refraction**.



Laws of refraction

1. The incident ray, Normal, refracted ray at the point of incidence all lie in the same plane
2. The ratio of sine of incidence angle to the sine of refracted angle is constant. **This is known as Snell's law**

Refractive Index

The refractive index of a material is the ratio of the speed of light in a vacuum to its speed in the medium. A vacuum is a region with no matter in it, not even air. However, the speed of light in air is very close to that in a vacuum. $n = \frac{c}{v}$, Where n= refractive index (no unit), c=speed of light in a vacuum =3x108m/s, v=speed of light in a given medium (m/s)

Snell's Law of refraction

The angles of incidence and refraction when light travels from one medium to another can be calculated using Snell's Law.

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Where n_1 = Refractive index of material 1, n_2 = Refractive index of material 2, θ_1 = Angle of incidence and θ_2 = Angle of refraction

Example 1: A light ray, with an angle of incidence of 35^0, passes from water to air. Find the angle of refraction using Snell's Law. Discuss the meaning of your answer. (refractive index of water is 1.33, for air it is 1).

Solution

According to Snell's Law:

$$\begin{aligned} n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\ 1{,}33 \sin 35^\circ &= 1 \sin \theta_2 \\ \sin \theta_2 &= 0{,}763 \\ \theta_2 &= 49{,}7^\circ. \end{aligned}$$

The light ray passes from a medium of high refractive index to one of low refractive index. Therefore, the light ray is bent away from the normal.

Example 2: A light ray passes from water to diamond with an angle of incidence of 75^0. Calculate the angle of refraction.

According to Snell's Law:

$$\begin{aligned} n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\ 1{,}33 \sin 75^\circ &= 2{,}42 \sin \theta_2 \\ \sin \theta_2 &= 0{,}531 \\ \theta_2 &= 32{,}1^\circ. \end{aligned}$$

Calculating the Critical Angle

Now we shall learn how to derive the value of the critical angle for two given media. The process is fairly simple and involves just the use of Snell's Law that we have already studied. Snell's Law states that:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_1 \sin \theta_c = n_2 \sin 90^0 \quad \text{(for the first diagram)}$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) \text{or } \theta_c = \sin^{-1}(n)$$

$$\text{Where } n = \frac{n_2}{n_1}$$

Example1: Given that the refractive indices of air and water are 1 and 1.33, respectively, find the critical angle.

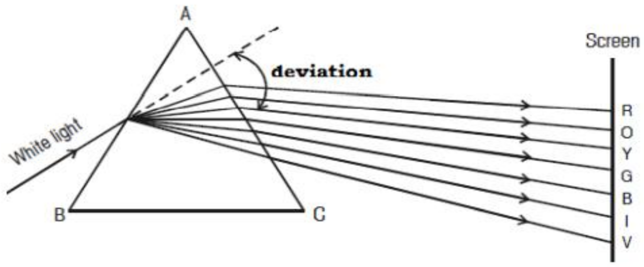
$$\begin{aligned} \theta_c &= \sin^{-1}\left(\frac{n_2}{n_1}\right) \\ &= \sin^{-1}\left(\frac{1}{1{,}33}\right) \\ &= 48{,}8^\circ \end{aligned}$$

Example2: Calculate the critical angle for glass-air interface, if the refractive index of glass is 1.50. **Answer: 41.8°**

Example3: Calculate the refractive index of diamond, if the critical angle for the diamond is 24°. **Answer: 2.46**

Dispersion of light by a prism

Dispersion is the splitting of white light into its constituent colours. This band of colours of light is called **its spectrum**.



Isaac newton observed that a beam of white light incident on a prism splits into its constituent colours to form "a visible spectrum. In the visible region of spectrum, the spectral lines are seen in the order from violet to red. The colours are given by the word VIBGYOR **(Violet, Indigo, Blue, Green, Yellow, Orange and Red) Violet colour** suffers the maximum deviation and **red the least**.

Refraction of light through thin lenses

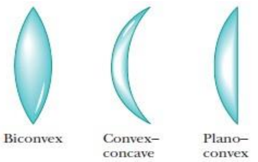
A **thin lens** is transparent medium usually made in glass or plastic bounded by one or two spherical surfaces.

1. Types and characteristics of lenses

Lenses are of two basic types, **convex** which are thicker than the edges and **concave** which the reverse is true.

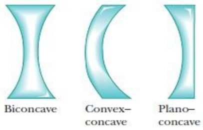
a. **Convex (Converging) lens**

Convex lens or **converging lens** is one which is thicker at the middle than at the center.



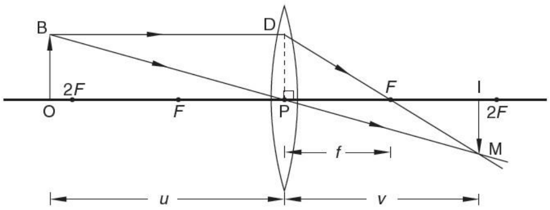
b. **Concave (Diverging) lens**

The **concave lens** or **diverging lens** is one which is thicker at the edges than at the middle



2. Lens formula

Consider a convex lens of focal length, *f*, which forms a real image IM of an object OB as shown in Fig. below



Triangles OBP and IMP are similar (3 angles are equal)

$$\therefore \frac{OB}{IM} = \frac{OP}{IP} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

Draw a line DP perpendicular to the principal axis where DP = BO. Triangles PDF and IMF are similar (3 angles are equal)

$$\therefore \frac{DP}{IM} = \frac{PF}{IF} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2)$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Magnification of the lens

$$M = \frac{v}{u} = \frac{h'}{h}$$

The position of image v is given by

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Where u =object distance

v =image distance

f =focal length

h =height of the object

h' = height of the object

Note: In calculation, the focal length of a diverging lens is taken as **negative**; the **image distance** *v* is also negative since the image is virtual.

Example

1. An object is placed i) 14cm, ii) 8cm in front of a convex lens of focal length10cm. Find the image distance and magnification in each case.

   Solution

   i) We have: u = +14cm (real object), f = +10cm (convex lens)

   Substitute in $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ or $\frac{1}{+10} = \frac{1}{+14} + \frac{1}{v}$

$$v = \frac{10 \times 14}{14 - 10} = \frac{140}{4} = 35$$

$the\ image\ distance\ is\ 35cm\ from\ the\ lens$

   The magnification: $m = \frac{35}{14} = 2.5$

   ii) We have d = 8cm (real object), f = +10 (convex lens)

   Substitute in, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

   $\frac{1}{+10} = \frac{1}{+8} + \frac{1}{v}$, $\frac{1}{v} = \frac{1}{10} - \frac{1}{8}$

   $v = \frac{8\times10}{8-10} = \frac{80}{-2} = -40$,

   $v = -40cm$, the minus sign means the image is virtual. Also the magnification

   $m = \frac{40}{8} = 5$

2. An object of height 2 cm is placed 8 cm from a convex lens and a virtual image is formed on the same side as the object at 24 cm from the lens. Calculate (a) the focal length of the lens (b) the height of the image formed.
3. A convex lens produces a real image of an object and the image is 3 times the size of the object. The distance between the object and the image is 80 cm. Calculate the focal length of the lens.

LO6: CHARACTERIZE SOURCES OF ENERGY IN THE WORLD

6.1. EXPLANATION OF BASIC CONCEPTS OF WORK, ENERGY AND POWER

- **Explanation of work**

Work is defined as the product of force and distance moved in the direction of the force. i.e. Work = force × distance moved in the direction of the force. $\mathbf{W = F \times d}$

The SI unit of work is **Joule**. Where 1 joule =1 newton × 1 meter

A joule is the work done when a force of 1 newton moves a body through a distance of 1 meter.

Bigger units used are: Kilojoules (1 kJ) = 1 000 J, Mega joule (1 MJ) = 1 000 000 J

Note: Whenever work is done, energy is transferred.

- **Positive work** when the direction of motion and that of the force are the same. For example, when a person is pushing a car, he does a positive work.
- **Negative work** when the direction of motion is opposite to the direction of the force. Examples; when a stone is thrown up vertically, the work of the force of gravity is negative.
- **The work is zero** when the displacement is zero despite of the action of the force. When a person tries to move a lorry and remains at rest, that person has done zero work.

Example1: A horizontal pulling force of 60 N is applied through a spring to a block on a frictionless table, causing the block to move by a distance of 3 m in the direction of the force. Find the work done by the force.

Example2: A horizontal force of 75 N is applied on a body on a frictionless surface. The body moves a horizontal distance of 9.6 m. Calculate the work done on the body.

- **Explanation of energy**

Energy is the ability or capacity to do work. Work done = Energy transferred. SI unit of energy is J**oules (J)**.

Mechanical energy: Mechanical energy is the energy possessed by a body due to its motion or due to its position. It can either be kinetic energy or potential energy of both. When an object is falling down through the air, it possesses both potential energy (PE) due to its position above the ground, and kinetic energy (KE) due to its speed as it falls. The sum of its PE and KE is its mechanical energy. **Mechanical energy = Kinetic energy + Potential energy.**

Potential energy: The energy possessed by a body (e.g. a stone) due to its position above the ground is called **gravitational potential energy**.

Example: A crane is used to lift a body of mass 30 kg through a vertical distance of 6.0 m. How much work is done on the body? What is the P.E stored in the body? Comment on the two answers.

Kinetic energy: It possesses energy as it moves. The energy which is possessed by a moving object due to its speed is called kinetic energy (KE). Examples of objects that possess KE include moving air, rotating windmills, falling water, rotating turbines and a moving stone. In general, any moving body possesses energy called **Kinetic energy.** The kinetic energy of a moving body is given by:

Kinetic energy = $\frac{1}{2}\mathbf{mv^2}$, where m and v are the mass and velocity of the body respectively.

Law of conservation of energy

If Initial P.E = Final K.E We say that energy has been conserved. This is summarized in the law of conservation of energy.

The law of conservation of energy states that energy cannot be created or destroyed but is simply converted from one form into another.

The law of conservation of mechanical energy

You should have learnt that the law of conservation of **mechanical energy states that**, the total mechanical energy (sum of potential energy and kinetic energy) in a closed system will remain constant/same (The sum of potential energy and kinetic energy anywhere during the motion must be equal to the sum of potential energy and kinetic energy anywhere else in the motion).

Work energy theorem

The work-energy theorem states that the work done on an object is equal to the change in its kinetic energy: $\Delta \boldsymbol{W} = \Delta \boldsymbol{K}.\boldsymbol{E} = \boldsymbol{K}.\boldsymbol{E}_{\boldsymbol{f}} - \boldsymbol{K}.\boldsymbol{E}_{\boldsymbol{i}}$

Example: The driver of a 1 000 kg car traveling at a speed of 16.7 m/s applies the car's brakes when he sees a red robot. The car's brakes provide a frictional force of 8000 N. Determine the stopping distance of the car.

- **Explanation of power**

Power is the rate of doing work. i.e. $Power = \frac{Work\ done}{Time\ taken} = \frac{ForceXDistance}{Time}$. The SI Unit of power is Watt (W) and $1watt = {}^{Joule}/_{second}$

Large units used are kilowatt and megawatt. 1Kilowatt = 1 000 W and 1Megawatt = 1 000 000 W

Example: What power is expended by a boy who lifts a 300N block through 10 m in 10s?

Answer: **Power=300Watt**

What is a kilowatt-hour (kWh)?

One kilowatt-hour (kWh) is the amount of energy used by an engine operating at a power of 1 kilowatt for 1 hour.$1kWh = 3.6*10^6\ J$

Example: Calculate the increase in kinetic energy of a car of mass 800 kg when it accelerates from 20 m s-1 to 30 ms^{-1}.

6.2: IDENTIFICATION OF TYPES OF ENERGY

- **Energy**

Energy is not visible; it occupies no space and has neither mass nor any other physical property that can describe it. However, it exists in many forms, some of these forms include:

Solar energy: This energy from the sun is in form of radiant heat and light.

Sound energy: In each case, kinetic energy has been converted to sound and heat energy**.**

Sound energy is the energy associated with the vibration or disturbance of bodies or matter.

Heat energy: The hotness is due to heat energy that has been transferred from the hot part to the cold part of the nail. Therefore, heat energy only travels from a hot object to a cooler one. Heat energy is a form of energy that is transferred from one body to another due to the difference in temperature.

Electrical energy: Electrical energy is the energy produced by the flow of electric charges (electrons).

Nuclear energy: Nuclear energy is the energy that results from nuclear reactions in the nucleus of an atom. It is released when the nuclei are combined or split.

Chemical energy: Chemical energy is a type of energy stored in the bonds of the atoms and molecules that make up a substance.

Source of energy

There are two kinds of energy sources;

1. Primary sources.

2. Secondary sources.

Primary sources of energy

Primary Sources are sources which can be used directly as they occur in the natural environment. They include:

- **Flowing water**: the flowing water from dams rotate turbines at the bottom of the dam which turn the generator resulting in generation of electricity
- **Nuclear**: Nuclear energy is created through reactions that involve the splitting or merging of the atoms of nuclei together
- **Sun:** The sun is the biggest source of energy and has played an important role in shaping our life on earth since the dawn of time.
- **Wind:** Wind is caused by the sun heating the earth unevenly. The air is heated differently causing hotter air to expand rise, and the colder one to condense and sink. This results to the movement of air and hence formation of wind.
- **Geothermal (interior of the earth**): Geothermal gradient is the difference in temperature between the core (interior) of the earth (planet) and its surface brings about conduction of heat from the core to the surface.
- **Fuels:** Fuels are substances which produce heat when burnt in the presence of oxygen.
- **Light energy:** The potential of light to perform work is called light energy. It is formed through chemical radiation and mechanical means. It is a form of energy produced by hot bodies and travels in a straight line.
- **Biomass (living thing and their waste materials):** Biomass is the total mass of organic matter in plant or animal. It is used to generate energy e.g. through burning to give heat energy.

Secondary sources of energy

Secondary sources are energy sources that are generated from primary sources. For instance, electricity is a secondary source because it is generated **for example** from solar energy using solar panels or from flowing water using the turbines to generate hydroelectricity.

Other secondary sources of energy include; petroleum products, manufactured solid fuels, gases, heat and bio fuel.

- **Identification of source of energy**

Renewable energy sources

A renewable energy source is an energy source which can’t be depleted/ exhausted. They exist infinitely i.e. never run out. They are renewed by natural processes.

Examples include: Sun, Geothermal, and Wind

Non-renewable energy sources

These are sources which can be depleted because they exist in fixed quantities. So they will run out one day.

Examples are: Coal, crude oil, natural gas, and uranium.

Fossil fuels like coal, crude oil, natural gas are mainly made up of carbon. They are usually found in one location because they are made through the same process and material. Millions of years ago dead sea organisms, plants, and animals settled on the ocean floor and in porous rocks. With time, sand, sediments and impermeable rock settle on the dead organic matter, as the matter continue to decay forming coal, oil and natural gas. Earth movements and rock shifts creates spaces that force these energy sources to collect at well-defined areas. With the help of technology, engineers are able to drill down into the sea bed to mine these sources and harness the energy stored in them.

Difference between Renewable and Non-renewable sources

No	Renewable energy sources	Non-renewable energy sources
1.	It can be used again and again throughout its life.	It cannot be used again and again but one day it will be exhausted.
2.	These are the energy resources which cannot be exhausted.	They are the energy resources which can be exhausted one day.
3.	It has low carbon emission and hence environment friendly.	It has high carbon emission and hence not environment friendly.
4.	It is present in unlimited quantity.	It is present in limited quantity and vanishes one day
5.	Cost is low.	Cost is high.
6.	Renewable energy resources are pollution free.	The non-renewable energy resources are not pollution free.
7.	It has high maintenance cost.	It has low maintenance cost as compared with the renewable energy resources.
8.	Large land area is required for the installation of its power plant.	Less land area is required for its power plant installation.
9.	Solar energy, wind energy, tidal energy etc are the examples of renewable resources.	Coal, petroleum, natural gases are the examples of non-renewable resources

Environmental effects of the use of energy sources

The following are some of the effects of use of the energy sources to the environment:

- Air and water pollution
- Deforestation
- Climate change and global warming.
- Land degradation

6.3. ANALYZING RELATIVE ADVANTAGES AND DISADVANTAGES OF VARIOUS ENERGY SOURCES

1. **Non- renewable (fossil fuel) energy**

Fossil fuels: They are remains of plant or animal that existed in a past geological age and that has been excavated from the soil. Types of fossil fuels

- **Coal** is a solid organic rock made up mostly of carbon. Coal was formed from the waste of plants that lived in forests and swamps millions of years ago.
- **Petroleum:** It is also called crude oil. It was produced after millions of years by the bacterial decomposition of animals and plants which were buried underground to great depths inside the earth due to the earth quakes, cyclones and storms.

Advantages of using fossil fuels

Fossil fuels are relatively cheap and easy to obtain. This is the reason why most people prefer using gas or kerosene for cooking over electricity.

Disadvantage of Burning fossil fuels

- Their supply is limited and they will eventually run out.
- Fossil fuels release carbon dioxide when they burn, which adds to the greenhouse effect and increases global warming. Of the three fossil fuels, for a given amount of energy released, coal produces the most carbon dioxide and natural gas produces the least.
- Coal and oil release Sulphur dioxide gas when they burn, which causes breathing problems for living creatures and contributes to acid rain

2. **Renewable sources**

I. **Wind energy**

Advantages of wind energy

- Very cheap way of generation of electricity
- Do not require fuels and can be run at low cost with minimum maintenance costs.
  - Safe because they do not produce substances which pollute air or water.

Disadvantages

- Very expensive to set up because of a lot of work must be done to select a proper site with correct wind velocity and abundant wind supply.
- The wind does not always blow, and wind direction may vary.

II. Geothermal energy

It is heat energy from the earth. Heat energy from the hot interior of the earth can rise the temperature of underground rocks in the earth‟s crust to very high temperatures

Advantages

- The pollution can be controlled by putting water or steam back into the earth High production in energy

Disadvantages

- The installation is very expensive; It requires a very greater space
- The sources are not easily accessible; there is loss of internal energy

III. Solar energy

Advantages

- The power source of the sun is absolutely free.
- Solar cells have no moving parts,
- The installation is not expensive
- No pollution of atmosphere

Disadvantages

- Used only in sun season; weak efficiency
- Storage system is expensive

IV. Hydropower

It is a transformation of the energy stored in a depth of water into electricity. The potential energy, or energy due to height, can be extracted by flowing the water through turbines as it moves from a higher level to a lower one.

Advantages

- The installation is not expensive: Big efficiency about 90%
- Source of water is available
- No pollution of atmosphere
- Require no fuel

Disadvantages

- Very expensive to set up because of high costs in setting suitable places, purchase of necessary materials and construction of the station and supply power lines.
- Some energy is lost in sound and internal energy.
- Weak power produced; its function is accorded to the seasons